

Class 5 — Immunity & Inflammation, Part 2: Autoimmunity & Hypersensitivity

NRSG 224 Pathophysiology · Fall 2026 · filled from the class study guide

Personal study sheet — not for submission.

Questions keep the guide's order. Brackets show where each answer came from: S = recorded-lecture slide, mm:ss = recording time (1x), sheet p. = your full study sheet. **not covered in lecture** = textbook knowledge, not from her lecture.

Sources: recorded lecture subtitles (times are the 1x recording, about ± 30 s; on the 2x video, halve them), recorded and class slides, verified clinical excerpts, and your full study sheet pp. 17-19.

Page 1 — Distinguishing self from nonself

Guide: "Self-tolerance (the absence of an immune response directed against a person's own antigens) normally is maintained through:" — two blank dashes, then MHC molecules (3rd mechanism, already printed).

A1 Self-tolerance mechanism #1 (blank -)

- Mechanism #1 = regulatory T cells (aka suppressor T cells) [02:13]
- Two jobs: shut off immune response once pathogen is gone; suppress any potentially self-reactive WBCs [02:13]
- Gets rid of those self-reactive cells before they can engage in autoimmune activity [02:48]
- Sheet version: turn off immune response + suppress self-reactive WBCs [sheet p.17]

A2 Self-tolerance mechanism #2 (blank -)

- Mechanism #2 = bone marrow + thymus screening of immature WBCs [02:48]
- T cells mature in thymus, B cells mature in bone marrow; both sites screen for potentially self-reactive cells [02:48]
- If found → get rid of self-reactive cells BEFORE release into general circulation [03:20]
- Sheet version: screen + remove immature self-reactive B cells and T cells [sheet p.17]

Page 1 — Inflammation

B1 S/S (cardinal signs): list all 5 (fills the five blank dashes) and for each, the mechanism she gave

- Redness + heat ← vasodilation → more blood to area [58:27]
- Swelling (edema) ← ↑ capillary permeability → fluid leaks into tissues [59:31]
- Pain ← swelling; usually goes along with it [59:31]
- Impaired/loss of function ← swollen parts don't work (e.g., sprained finger can't bend) [1:00:02]
- Cardinal signs listed: redness, heat, swelling, pain, impaired function [sheet p.18]

B2 Vascular response: momentary vasoconstriction, then vasodilation → ____

- Momentary vasoconstriction → vasodilation [sheet p.18]
- Vasodilation = first vascular event → brings more blood: O₂, fuel, antibodies to fight

infection [58:27]

- Result = the redness and warmth of inflammation [58:59]

B3 ↑ capillary permeability → fluid escapes into tissues → ____

- ↑ capillary permeability = capillaries get 'leaky' → fluid leaks into tissues, bringing nutrients + antibodies [59:31]
- Causes edema, pain, and loss of function [59:31]
- Exudate types: serous (clear, light yellow = serum), serosanguineous (watery blood/pink), purulent (pus = dead WBCs, infected) [1:01:08]; [slide 53]
- Exudate is usually GOOD: brings nutrients/O₂/antibodies, flushes dirt, dilutes irritants, slows spread of contaminants [1:02:13]

B4 Cellular response - movement of WBCs into area of injury → ____

- Cellular response = phagocytic WBCs migrate to the injury area [1:03:53]
- Neutrophils get there first, followed by monocyte-macrophages [1:03:53]
- They phagocytize pathogens + dead tissue → get it ready for repair [1:03:53]
- If needed, they act as APCs → signal/activate adaptive immunity [1:04:24]

B5 Acute-phase response (fever, ↑ # WBCs, skeletal muscle catabolism) — what to know

- Acute-phase response = systemic effects: fever, leukocytosis, ↑ inflammatory markers, varying degrees of skeletal muscle catabolism [slide 57]; [1:05:39]
- Fever is beneficial: bacteria can't survive higher temp + WBCs become more active [1:05:26]
- Leukocytosis = actual ↑ in number of WBCs; fever + ↑ WBC ≠ always infection (post-op incision, MI) → monitor trends, check pus, wound culture [1:05:39]
- Skeletal muscle catabolism = break down muscle protein → amino acids to make antibodies / repair tissue; body mobilizes only what's useful (mosquito bite vs appendicitis) [1:10:21]
- Markers: CRP (liver; regulates cytokines), fibrinogen (helps locate pathogens), ESR (markers stick to RBCs → heavier → sediment faster) [36:36]; [37:37]
- ★ **Prof emphasis:** 'those are the only three I'm going to expect you to recognize for the test' (CRP, fibrinogen, ESR) [39:49]

B6 Potential problems r/t inflammation

- Inflammation can get out of control: very severe, or recurrent/persistent → harmful [1:12:01]
- Chronic irritation/inflammation → metaplasia + scar tissue formation [1:12:34]

- Scar tissue → loss of function + deformity (RA joints pulled into abnormal positions) [1:13:09]
- Kidney scarring → can't filter blood / make urine like real kidney tissue [1:13:41]

B7 Potential problems if immune system function impaired/altered

- Immunodeficiency → infection [class slide 4]
- Hypersensitivity → allergic reactions → unnecessary inflammation [class slide 4]
- Autoimmunity → damage to self tissue [class slide 4]
- Diagnostics: immunodeficiency → cultures, WBC count; hypersensitivity → allergy testing; autoimmunity → check for autoantibodies [class slide 5]
- ★ **Prof emphasis:** ALL three can cause inflammatory markers to increase [class slide 5]

Flowchart 1 — Autoimmunity (breakdown in self-tolerance)

C1 Nursing problem:

- Nursing problem: damage to self tissue — immune system attacks self-antigens [class slide 4]
- Injured self cells → organ/body system not working properly [22:17]
- Most autoimmune diseases are chronic: cycles of remissions and exacerbations/flares [32:40]
- Goal of care: minimize exacerbations, get patients into remission, keep them there as long as possible [33:14]

C2 SLE — Assessment findings:

- Ag-Ab complexes deposit in collagen-rich tissues → many body systems; skin: bruising, rashes, classic butterfly/malar rash (worsened by sun; flares triggered by sun) [46:46]; [47:19]
- Joints (lupus arthritis), kidney (inflammation, renal failure), lungs, heart, eyes, GI tract; not all systems in every patient [48:01]
- Autoantibodies: ANA (anti-nuclear) is a primary one — antibodies vs nuclear/DNA antigens; also autoantibodies vs RBCs and platelets [43:32]; [45:34]
- Labs: ↑ inflammatory markers (CRP, fibrinogen, ESR) = inflammation only; autoantibody list on slide also Anti-DNA, Anti-CCP, Anti-phospholipids, RF [slide 34]; [35:28]
- Dx is a diagnosis of exclusion — 'the great imitator'; no one definitive test; can take years [48:43]

C3 Interventions (autoimmune / SLE):

- Two levels of intervention: stop the underlying autoimmune attack + treat the symptoms [41:39]; [slide 35]
- Stop attack: immunomodulators — immunosuppressive drugs, monoclonal antibodies, corticosteroids; also plasmapheresis [41:58]; [slide 44]
- Treat symptoms/inflammation: NSAIDs (Motrin, ibuprofen, aspirin) — fewer side effects, less powerful; corticosteroids = synthetic cortisol, powerful but many side effects [51:28]

- Use NSAIDs first, add corticosteroids if necessary (NSAIDs won't halt kidney damage) [52:31]
- Keep lupus patients out of the sun — sun exposure triggers exacerbations [47:19]

Flowchart 2 — Hypersensitivity disorders

D1 Nursing problem:

- Problem: immune system reacts to an antigen most people would not react to; not normal/appropriate, usually excessive → can damage host tissue [1:14:17]
- Results in unnecessary inflammation / allergic reactions [class slide 4]
- Hypersensitivity = broad umbrella; allergy (reacts to environmental antigen, e.g., cat dander) and autoimmune disease (reacts to self-antigen) are two specific types [1:14:50]
- Predisposing factors: BOTH genetic predisposition + exposure to triggering antigen (inhaled, ingested, injected, skin contact) [1:16:20]

D2 Type I — allergic rhinitis (hay fever)

- Type I = IgE-mediated; allergic rhinitis = 'hay fever' [1:24:49]
- Findings: runny nose, lots of sneezing, red/itchy/watery eyes [1:24:49]
- Mechanism: mast cells/basophils degranulate in nasal passages / eye mucous membranes [1:27:27]

D3 Type I — asthma

- Type I, IgE-mediated; airway constriction [1:25:12]
- Findings: wheezing, shortness of breath [1:25:12]
- Mechanism: degranulation in the airways [1:27:27]
- Reaction within ~15 min of trigger exposure (e.g., cat-allergic person starts to wheeze in ~15 min) [1:37:56]

D4 Type I — urticaria

- Urticaria = hives; Type I, IgE-mediated [1:25:43]
- Looks like mosquito bites all over the body; very itchy [1:25:43]
- Mechanism: degranulation in the skin [1:27:27]

D5 Type I — atopic dermatitis (eczema)

- Atopic dermatitis = eczema; Type I, IgE-mediated [1:25:12]
- Findings: itchy, scaly, crusty rashes — 'incredibly itchy' [1:25:12]
- Mechanism: degranulation in the skin [1:27:27]

D6 Type I — anaphylaxis

- Most severe Type I disorder; main worry = blood pressure drops → hypotension → can progress to anaphylactic shock; people can die [1:26:14]
- Also bronchoconstriction and some shortness of breath [1:26:14]
- Mechanism: degranulation system-wide → whole-body vasodilation → BP drop [1:27:48]

- Related: angioedema = tongue, lips, part of face swell [1:25:43]
- Textbook, not from her lecture: anaphylaxis emergency care = IM epinephrine first, then airway/oxygen, IV fluids for hypotension, plus antihistamines/corticosteroids

[not covered in lecture — textbook]

not covered in lecture

D7 Type II — blood transfusion reactions

- Type II = antibody-mediated/cytotoxic; uses IgG and IgM (not IgE) [1:30:28]
- Example: type A recipient (has antibodies vs B antigen) given type B blood → recipient antibodies attack the donor cells [1:31:00]
- When antibody binds antigen on donor cell → destroys the donated cell; the binding itself causes the damage [1:31:00]

D8 Type III — SLE (why it is Type III)

- Type III = immune complex-mediated (IgG/IgM); immune complex = antigen-antibody complex [1:32:10]
- Not the binding itself — complexes get deposited in tissue → activate complement → inflammation = the symptoms [1:33:09]
- That is exactly the lupus scenario (ANA binds DNA antigen → complexes deposit in collagen-rich tissue) [44:33]; [1:34:13]

D9 Type IV — contact dermatitis (poison ivy), diabetes mellitus

- Type IV = T-cell-mediated / cellular — only type NOT antibody-mediated; delayed [1:34:44]

- T cells get sensitized to antigen → next exposure, T cells attack → symptoms [1:34:44]
- Examples: poison ivy/poison oak contact dermatitis (rash where touched); type 1 diabetes (sensitized T cells attack pancreatic cells); certain transplant rejections [1:35:16]
- Delayed timing: rash/peak ~1-3 days after exposure (vs Type I/II 15-30 min, Type III ~6 hr) [1:37:56]

D10 Assessment findings (hypersensitivity) — guide says 'Specific to individual disorders'

- Findings are specific to individual disorders; symptoms differ by WHERE mast cells/basophils degranulate (airways→asthma, nose/eyes→rhinitis, skin→eczema/hives, system-wide→anaphylaxis) [1:27:15]
- Diagnostics: allergy testing — skin testing, sticking little bits of antigen under the skin to see which cause a reaction [1:29:22]; [class slide 5]
- All three major immune problems can increase inflammatory markers [class slide 5]
- ★ **Prof emphasis:** on the test she can't ask Type II vs Type IV from symptoms; but anything that sounds like an 'allergy' = Type I IgE-mediated [1:36:33]

D11 Interventions (hypersensitivity):

- Main interventions: get rid of/avoid the trigger + treat the symptoms; we can't desensitize once a person is sensitized [1:39:01]
- Meds for Type I: antihistamines (histamine is a mediator) + anti-inflammatories —

corticosteroids local vs systemic (e.g., fluticasone), others [1:29:05]; [slide 70]

- Help patients identify allergens (skin testing) and avoid them; if unavoidable (common trees, pets) → rely more on medications
- ★ **Prof emphasis:** we can NEVER stop monitoring patients for allergic reactions to meds — B cells may start making IgE after the 1st, 10th, or 50th exposure (penicillin story)

[1:23:32]